

Cara

Software Requirements Specification — Postnatal Follow-Up Coordination Platform

Health-a-thon 2026 | Maternal & Women’s Health Track

September 2026

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1. Document Control

Field	Value
Project Name	Cara
Version	1.0
Date	September 2026
Document Purpose	Implementation-ready SRS for the Health-a-thon 2026 (Koita Foundation / KCDH-IIT Bombay / FOGSI) Maternal & Child Health track build sprint
Prepared By	[Team name]
Intended Audience	Hackathon mentors/judges, developers, doctor co-founder, QA
Revision History	v1.0 — initial SRS drafted from idea-submission solution

2. Introduction

2.1 Purpose

Cara tracks whether a mother and newborn complete the postnatal follow-up visits a doctor scheduled at hospital discharge, and closes the loop with automated, non-clinical WhatsApp/SMS reminders to the family and a ranked worklist for the facility's own nursing staff.

2.2 Scope

In scope: structured discharge-plan entry by a doctor; calendar computation for the four WHO-defined postnatal visits (24h, 48–72h, 7–14 days, 6 weeks) plus doctor-set risk-based interval overrides; a coordinator-facing worklist ranked by days overdue; automated WhatsApp/SMS nudges to the mother and one consented family member; a due-vs-completed outcomes summary; an immutable audit log.

Out of scope: any diagnostic, triage, or treatment-recommendation logic; OCR/free-text parsing of discharge summaries; integration with RCH Portal, ANMOL, or Kilkari (positioned as complementary, not integrated, for MVP); a patient-facing mobile app (mother/family interact only via WhatsApp/SMS); real patient data of any kind.

Target environment: a single facility or small clinic network's own staff (not a government-scale, multi-facility rollout) for the hackathon MVP.

2.3 Problem Definition

A doctor's postnatal follow-up plan, once the patient is discharged, has no software system tracking whether it is actually completed. Cara converts that plan into machine-tracked calendar entries, compares scheduled dates to actual completion, and automates the reminder and worklist work a facility currently does manually or not at all.

2.4 Objectives

- O1: Reduce time to detect a missed postnatal visit from “never noticed” to same-day, via an automatically ranked worklist.
- O2: Deliver at least one reminder to the mother/family for every scheduled visit without manual staff effort.
- O3: Produce a facility-level due-vs-completed report with zero manual data compilation.
- O4: Guarantee zero clinical-content generation — verified by an automated content filter on every AI-generated message.

2.5 Intended Users

User Type	Role
Doctor / OBGYN	Sets the discharge plan and any risk-based visit-interval overrides once per patient
Coordinator (nurse / lactation counselor)	Owens the day-to-day worklist; records visit completion; manages family consent
Mother	Receives WhatsApp/SMS nudges; no login required
Family member	Opt-in, consented second recipient of nudges; no login required
Facility Admin	Manages staff accounts and facility configuration

2.6 Definitions and Acronyms

PNC — Postnatal Care. FOGSI — Federation of Obstetric and Gynaecological Societies of India. WHO — World Health Organization. RBAC — Role-Based Access Control. IVR — Interactive Voice Response. PII — Personally Identifiable Information.

2.7 References

WHO postnatal care visit guidelines (24h / 48-72h / 7-14 day / 6-week schedule); FOGSI clinical standards and Gestosis score for risk-based follow-up intervals; India MoHFW Guidance Note on Optimizing Post-Natal Care (2022); Twilio WhatsApp Business API / SMS sandbox documentation.

3. System Overview

Cara has three surfaces: a staff web dashboard (doctor + coordinator + admin), an outbound WhatsApp/SMS channel (mother + family, no login), and a scheduled background engine that computes due/overdue status daily and triggers nudges. Data flows one way from the doctor's one-time plan into the calendar, and the coordinator's completion entries flow back into the same calendar — no other write path exists.

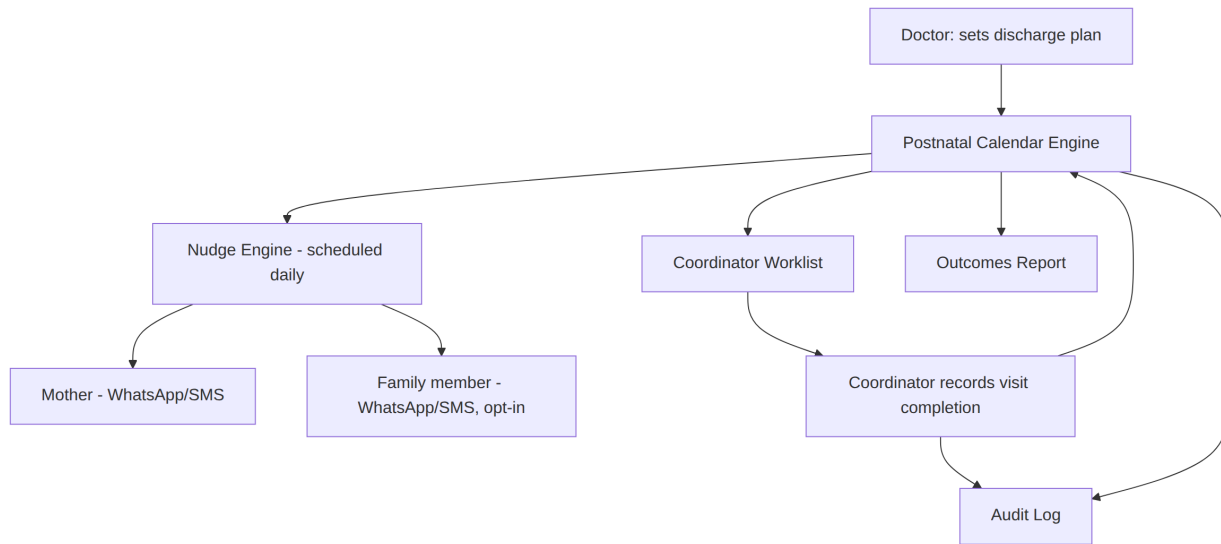


Figure 1: System Overview: Data Flow Between Doctor, Calendar, Worklist, and Nudges

Major inputs: discharge plan (visit dates, risk overrides), visit-completion entries, consent toggles. **Major outputs:** worklist view, WhatsApp/SMS nudges, outcomes report, audit trail. **Key workflow:** Discharge Plan Creation → Calendar Computation → Worklist/Nudge Cycle → Visit Completion → Report.

4. User Roles & Actors

Actor	Description	Permissions	Main Actions
Doctor	Sets/reviews discharge plan and risk overrides	Create/edit discharge plan; view worklist and reports (read-only)	Set visit schedule, apply risk-based interval override
Coordinator	Nurse or lactation counselor at the facility	Full access to worklist, visit completion, consent management	Mark visit complete/missed, add family member, review digest
Facility Admin	Manages accounts	Create/deactivate staff accounts, configure facility	Staff onboarding, facility settings
Mother	Discharge patient	None (no login)	Receives nudges; responds via WhatsApp if two-way is enabled
Family member	Consented secondary recipient	None (no login)	Receives nudges
Nudge Engine (automated)	Scheduled background process	System-level	Computes overdue status daily; triggers messages via Twilio
Twilio (external)	Messaging provider	API-key scoped	Delivers WhatsApp/SMS; returns delivery-status webhooks

5. Functional Requirements

Module 1 — Discharge Planning

FR-001 — Create Discharge Plan - Description: Doctor creates a postnatal follow-up plan for a newly discharged patient. - Actor: Doctor - Preconditions: Patient record exists (or is created in the same flow); doctor is authenticated. - Trigger: Doctor submits the discharge-plan form. - Input: Patient identifier, delivery date, discharge date, mother's contact number, preferred language. - Processing: System generates four default visit records (24h, 48-72h, 7-14d, 6wk) from delivery date using WHO intervals. - Output: Discharge plan record with four scheduled PostnatalVisit entries. - Postconditions: Calendar entries exist and are visible on the coordinator worklist. - Alternative flow: Doctor edits a default visit date before saving (e.g., delivery complications delaying discharge). - Exception: Missing mandatory contact number → form blocks submission with inline error. - Priority: Must

FR-002 — Apply Risk-Based Interval Override - Description: Doctor flags a risk condition (e.g., hypertensive disorder) that requires closer follow-up per FOGSI guidance. - Actor: Doctor - Preconditions: Discharge plan exists. - Trigger: Doctor selects a risk flag from a fixed list during or after plan creation. - Input: Risk flag type (hypertension, postpartum hemorrhage history, anemia, C-section, other — free-text disabled to avoid clinical-content drift). - Processing: System applies a pre-configured interval rule (e.g., weekly visit added for 4 weeks) instead of/in addition to the default four visits. - Output: Updated visit schedule reflecting the risk-based cadence. - Postconditions: Worklist and nudges reflect the new schedule. - Alternative flow: Doctor removes a risk flag before discharge is finalized. - Exception: Two conflicting risk flags selected → system takes the more frequent cadence, does not average or infer. - Priority: Must

FR-003 — Edit Discharge Plan Post-Creation - Description: Doctor amends a visit date after initial creation (e.g., patient readmitted). - Actor: Doctor - Priority: Should

FR-004 — Deactivate/Close Discharge Plan - Description: Doctor or coordinator closes a plan once all visits are complete or the patient transfers care elsewhere. - Actor: Doctor, Coordinator - Priority: Should

Module 2 — Postnatal Calendar Engine

FR-005 — Compute Visit Status - Description: Daily job computes each visit's status as Upcoming, Due Today, Overdue (with days-overdue count), or Completed. - Actor: System (scheduled job) - Preconditions: At least one active discharge plan exists. - Trigger: Daily cron run (00:05 local time). - Input: All active PostnatalVisit rows. - Processing: Compare current date to due date and completion status. - Output: Updated status field per visit. - Postconditions: Worklist and nudge engine read the refreshed status. - Exception: Job failure → retried once after 15 minutes; alert logged if second attempt fails. - Priority: Must

FR-006 — Record Visit Completion - Description: Coordinator marks a visit as completed after contact with the patient. - Actor: Coordinator - Input: Visit ID, completion date, optional free-text note limited to non-clinical logistics (e.g., "seen at home visit") — note field is explicitly rules-filtered to reject clinical terminology. - Output: Visit status set to Completed; audit entry created. - Priority: Must

FR-007 — Record Visit as Missed (Unreachable) - Description: Coordinator logs that a visit could not be completed and the reason is operational (unreachable, declined, moved). - Actor: Coordinator - Priority: Should

Module 3 — Coordinator Worklist

FR-008 — View Ranked Worklist - Description: Coordinator sees all active patients at their facility sorted by days-overdue descending. - Actor: Coordinator - Input: Facility ID (from session). - Output: List of patient/visit pairs with days-overdue, contact info, and a one-click “mark complete” action. - Priority: Must

FR-009 — Filter Worklist - Description: Coordinator filters the worklist by visit type or risk flag. - Actor: Coordinator - Priority: Should

FR-010 — Patient Detail View - Description: Coordinator opens a single patient’s full visit history and consent status. - Actor: Coordinator - Priority: Must

Module 4 — Nudge & Communication Engine

FR-011 — Trigger Scheduled Nudge - Description: System sends a WhatsApp (falling back to SMS on delivery failure) message when a visit becomes Due Today or crosses into Overdue. - Actor: System (Nudge Engine) - Preconditions: Consent on file for the recipient; visit status changed to Due Today or newly Overdue. - Input: Visit record, recipient contact, preferred language. - Processing: Populate a template (visit type, due/overdue date) via Twilio API; if a multilingual AI platform is available, generate the message text in the recipient’s preferred language and pass it through the content filter (see AI-006) before send. - Output: Message sent; delivery attempt logged. - Postconditions: NudgeLog entry created. - Alternative flow: WhatsApp delivery fails → automatic SMS fallback. - Exception: No valid contact number → nudge skipped, flagged on worklist for manual follow-up. - Priority: Must

FR-012 — Generate Weekly Plain-Language Digest - Description: System compiles a coordinator-facing weekly summary of due/overdue counts per patient panel. - Actor: System, reviewed by Coordinator - Priority: Should

FR-013 — Twilio Delivery-Status Webhook Handling - Description: System receives and records delivery status (sent/delivered/failed) from Twilio. - Actor: System - Priority: Must

FR-014 — Manual Nudge Resend - Description: Coordinator manually re-triggers a nudge for a specific patient. - Actor: Coordinator - Priority: Could

FR-015 — Two-Way Reply Capture (Optional) - Description: If the mother replies to a WhatsApp nudge, the reply is logged for coordinator review (not auto-interpreted). - Actor: System, Coordinator - Priority: Could

Module 5 — Consent & Family Access

FR-016 — Add Family Member with Consent Capture - Description: Coordinator (with mother’s verbal/written consent, recorded at intake) adds a family member’s contact for opt-in nudges. - Actor: Coordinator - Input: Family member name, relation, phone number, consent confirmation flag. - Output: FamilyMember and ConsentRecord created. - Priority: Must

FR-017 — Revoke Consent - Description: Mother or coordinator revokes a family member’s access at any time. - Actor: Coordinator, (on behalf of) Mother - Output: Family member’s nudges stop immediately; audit entry created. - Priority: Must

FR-018 — Consent Status Visibility - Description: Patient detail view always shows current consent status for any linked family member. - Actor: Coordinator - Priority: Should

Module 6 — Outcomes Reporting

FR-019 — Generate Due-vs-Completed Report - Description: Facility admin/doctor views aggregate completion rate by visit type over a selected date range. - Actor: Facility Admin,

Doctor - Output: Report screen and CSV export. - Priority: Should

FR-020 — Export Anonymised Summary - Description: Report can be exported with patient identifiers stripped, for external sharing (e.g., to a funder or accreditation reviewer). - Actor: Facility Admin - Priority: Could

Module 7 — Audit & Compliance Logging

FR-021 — Immutable Audit Log - Description: Every calendar mutation, consent change, and nudge send is written to an append-only audit log with timestamp and actor ID. - Actor: System - Priority: Must

Module 8 — Facility & Staff Administration

FR-022 — Staff Account Management - Description: Facility admin creates, edits, and deactivates staff accounts with role assignment (Doctor / Coordinator / Admin). - Actor: Facility Admin - Priority: Must

FR-023 — Facility Configuration - Description: Admin sets facility name, default visit-interval templates, and enabled languages. - Actor: Facility Admin - Priority: Should

6. User Stories

US-001 — As a doctor, I want to set a patient’s postnatal visit schedule at discharge in under two minutes, so that follow-up doesn’t depend on memory. *Given* a newly delivered patient, *when* I submit the discharge form, *then* four visit records are created and visible on the coordinator worklist.

US-002 — As a doctor, I want to flag a high-risk patient for closer follow-up, so that her visit cadence matches FOGSI guidance. *Given* a discharge plan, *when* I select a risk flag, *then* the visit schedule updates to the corresponding interval.

US-003 — As a coordinator, I want to see which of my patients are overdue, ranked by severity, so that I know who to contact first. *Given* an active patient panel, *when* I open the worklist, *then* I see all overdue patients sorted by days-overdue.

US-004 — As a coordinator, I want to mark a visit complete after contacting the patient, so that the calendar and reports stay accurate. *Given* a due visit, *when* I record completion, *then* the visit status updates and the audit log records the change.

US-005 — As a mother, I want a reminder in my own language when a check-up is due, so that I don’t miss it. *Given* a visit due today, *when* the nudge engine runs, *then* I receive a WhatsApp or SMS message in my preferred language.

US-006 — As a family member, I want to be included in reminders (with the mother’s consent), so that I can help her keep appointments. *Given* recorded consent, *when* a nudge is sent to the mother, *then* I receive the same reminder.

US-007 — As a facility admin, I want a due-vs-completed report, so that I can show our follow-up outcomes to a funder or accreditation reviewer. *Given* a date range, *when* I generate the report, *then* I see completion rates per visit type with an anonymised export option.

US-008 — As a coordinator, I want to revoke a family member’s access immediately if the mother asks, so that consent stays meaningful. *Given* an active family-member link, *when* I revoke consent, *then* nudges to that contact stop immediately.

7. Use Cases

UC-01 — Create Discharge Plan Primary Actor: Doctor. Secondary: System. Goal: Establish a trackable postnatal calendar at discharge. Preconditions: Patient registered. Main Flow: Doctor opens discharge form → enters delivery/discharge dates and contact → optionally applies risk flag → submits → system generates visit records. Alternative Flow: Doctor edits a generated date before saving. Exception Flow: Missing contact number blocks submission. Postconditions: Active calendar exists.

UC-02 — Complete a Visit Primary Actor: Coordinator. Goal: Reflect a real-world visit in the system. Main Flow: Coordinator opens worklist → selects patient → marks visit complete with date → system updates status and audit log.

UC-03 — Automated Nudge Delivery Primary Actor: System (Nudge Engine). Secondary: Twilio. Goal: Remind mother/family without manual staff action. Main Flow: Daily job flags Due Today/Overdue visits → engine composes message → sends via Twilio → logs delivery status. Exception Flow: Delivery fails → SMS fallback attempted → if still failed, worklist flags for manual contact.

UC-04 — Grant Family Consent Primary Actor: Coordinator. Secondary: Mother (verbal consent captured by coordinator). Goal: Add a second nudge recipient with documented consent. Main Flow: Coordinator adds family member details and confirms consent was given → system activates nudges for that contact.

UC-05 — Generate Outcomes Report Primary Actor: Facility Admin. Goal: Produce a due-vs-completed summary for reporting. Main Flow: Admin selects date range → system aggregates visit statuses → report rendered with export option.

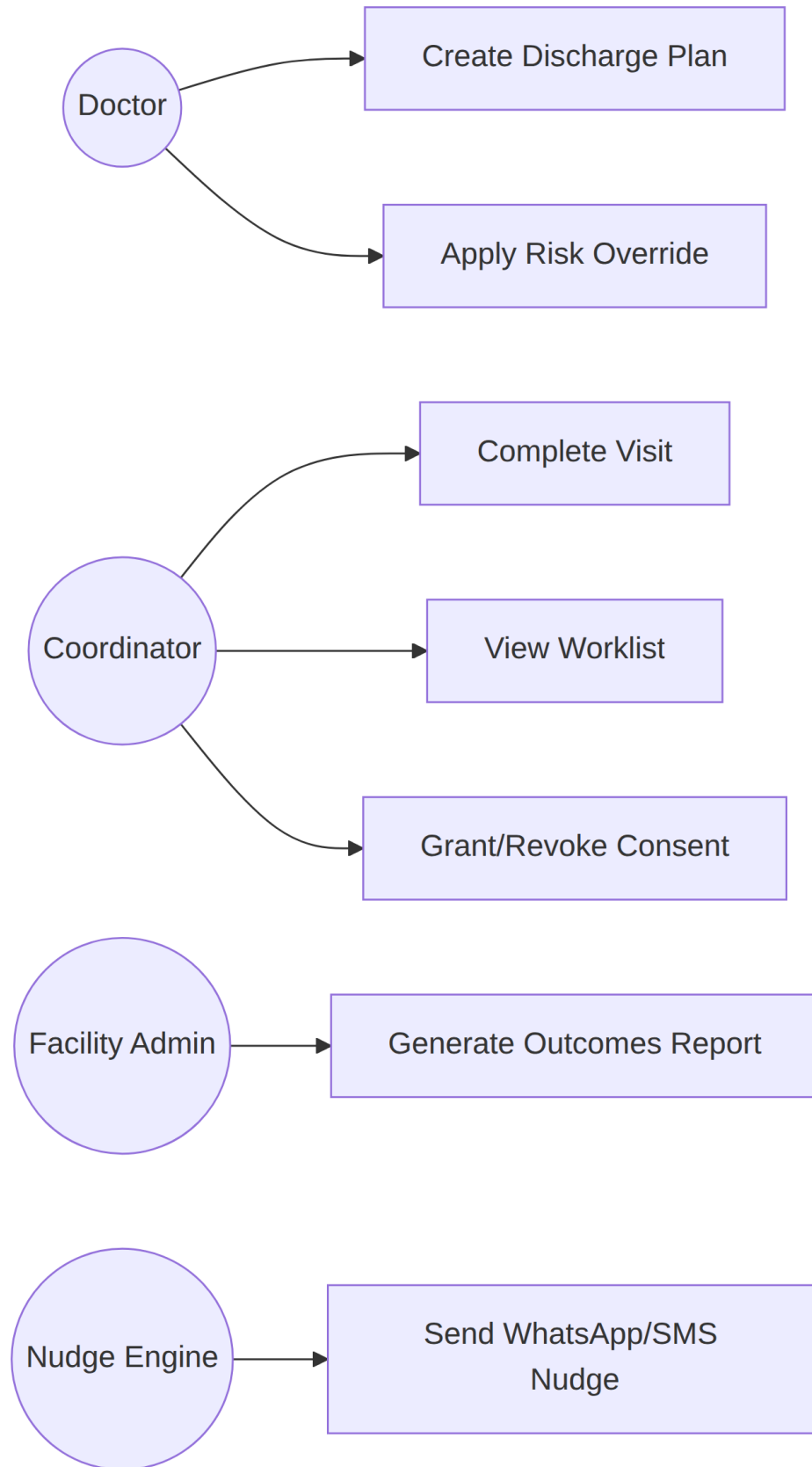


Figure 2: Use Case Diagram: Actors and Their Primary Interactions

8. System Workflows

Workflow A — Discharge Plan Creation Input (doctor form) → Validation (mandatory contact/date fields) → Processing (generate 4 default visits, apply risk override if any) → Decision (any risk flag?) → Action (write visit records) → Output (calendar populated) → Storage (Postgres) → Feedback (confirmation screen to doctor).

Workflow B — Nudge Delivery Cycle

Workflow C — Visit Completion & Calendar Update Input (coordinator action) → Validation (visit belongs to coordinator's facility) → Processing (update status, timestamp) → Storage (Postgres) → Audit (append-only log entry) → Feedback (worklist refreshes, patient removed from overdue list).

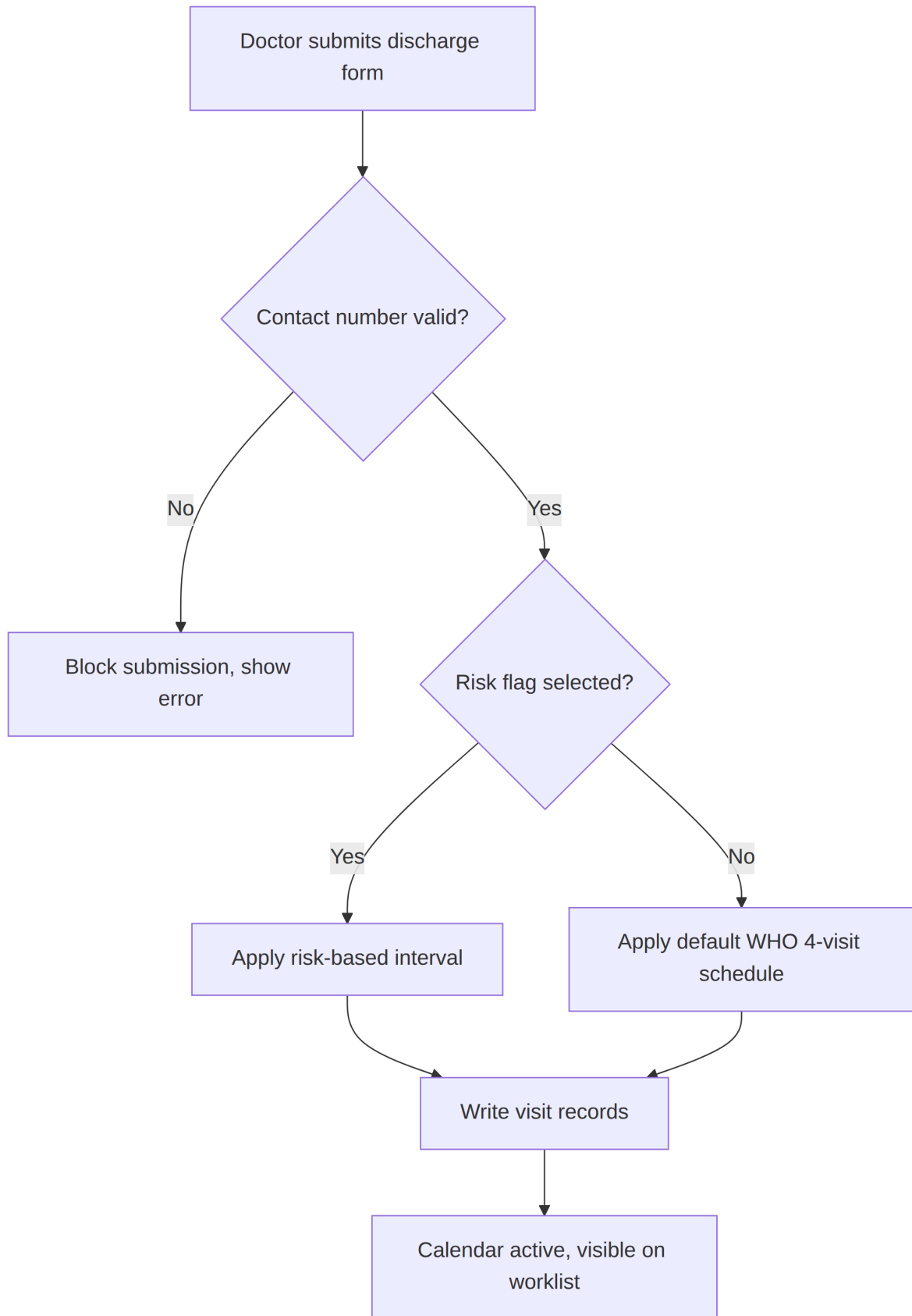


Figure 3: Workflow A: Discharge Plan Creation

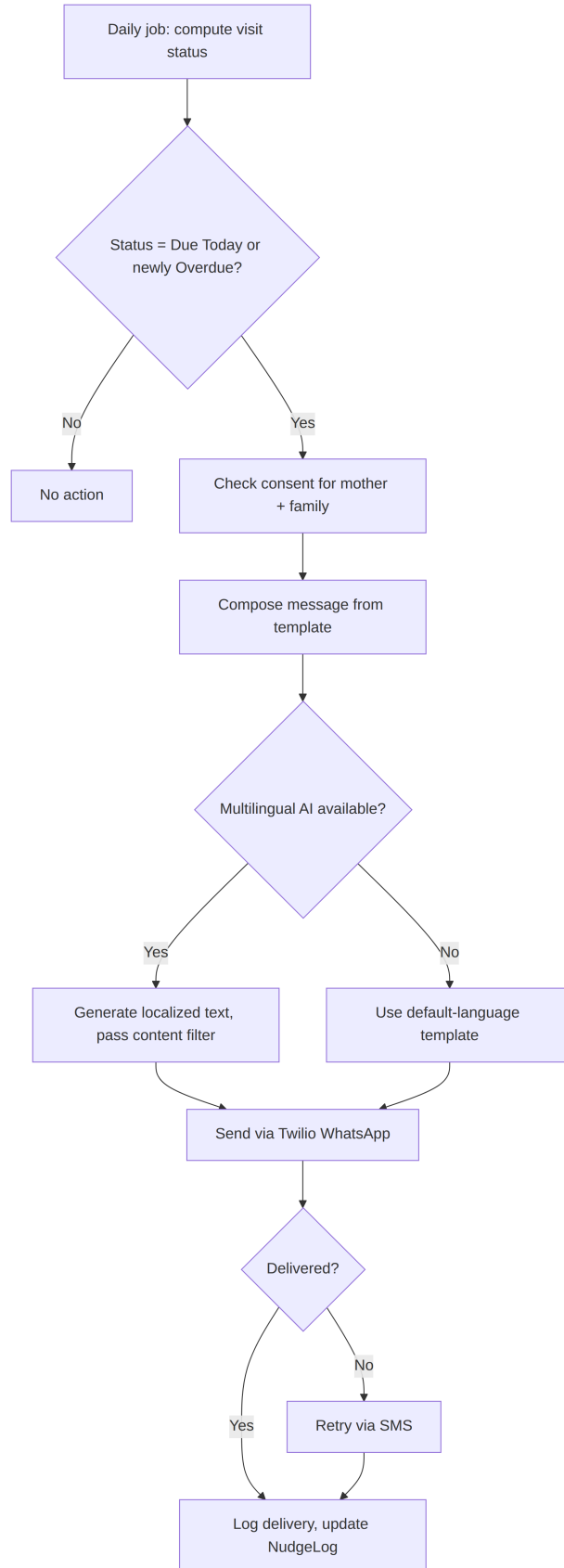


Figure 4: Workflow B: Nudge Delivery Cycle

9. Non-Functional Requirements

Performance: Worklist for a panel of up to 500 active patients loads in under 2 seconds. Nudge batch job for up to 1,000 due/overdue visits completes within 5 minutes.

Scalability: MVP designed for a single facility (hundreds of patients); architecture (stateless API + Postgres + queue) allows horizontal scaling of the API layer without redesign for a future multi-facility rollout.

Availability: Hackathon prototype target 95% uptime during demo/pilot window; no formal SLA required for MVP.

Reliability: Failed nudge sends are retried once automatically; failures beyond that are surfaced on the worklist, never silently dropped.

Security: JWT-based staff authentication; RBAC enforced at the API layer; all traffic over HTTPS; Twilio webhook signatures verified.

Privacy: All patient/family contact data encrypted at rest; consent is mandatory and revocable before any family-member nudge is sent; MVP demo/testing uses only synthetic data.

Usability: No app install required for mother/family (WhatsApp/SMS only); staff dashboard usable on a standard laptop browser, no specialized hardware.

Maintainability: Modular backend (one service per module in Section 5); structured logging with request IDs; no module directly writes to another module's tables.

Portability: Backend containerized (Docker); deployable to any standard cloud VM or PaaS without vendor-specific dependencies beyond Twilio.

10. System Architecture Requirements

Component	Responsibility	Input	Output	Dependencies
Frontend (staff dashboard)	Doctor/coordinator/admin UI	User actions	API calls	Backend API
Backend API (FastAPI)	Business logic, auth, RBAC	HTTP requests	JSON responses	Postgres, Redis
Postgres	Persistent storage	SQL writes/reads	Data	—
Scheduler (APScheduler/Celery + Redis)	Daily status computation, nudge triggering	Cron trigger	Job execution	Backend, Postgres
Twilio Integration	WhatsApp/SMS delivery	Message payload	Delivery status webhook	External Twilio account
Multilingual LLM (optional)	Digest/nudge text generation in preferred language	Structured due/overdue data	Localized text	External LLM API, content filter
Content Filter	Rejects any clinical-content or off-template output before send	Generated text	Pass/fail decision	Rules engine (regex/keyword list, no ML)
Audit Log Store	Immutable event history	Mutation events	Append-only records	Postgres (separate table, insert-only)

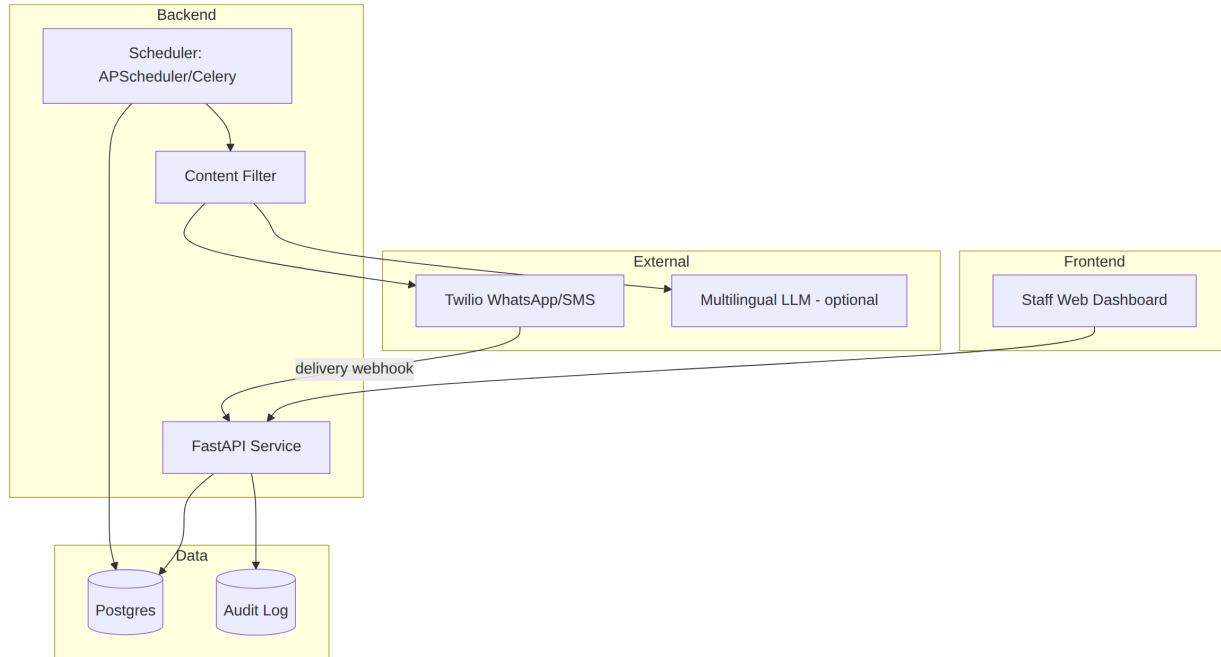


Figure 5: System Architecture: Components and Dependencies

11. AI / ML Requirements

AI-001 — Model Purpose: Generate plain-language nudge text and a weekly coordinator digest from structured due/overdue data, and translate that text into the recipient’s preferred language. The model never predicts risk, diagnoses, or recommends care.

AI-002 — Inputs: Visit type, days overdue/due, target language, minimal patient identifier (first name or initial only).

AI-003 — Processing: Fixed prompt template constrains output to scheduling language only; no open-ended generation.

AI-004 — Output: Short message (under ~300 characters) for WhatsApp/SMS, or a short weekly digest paragraph for the coordinator.

AI-005 — Performance: No accuracy/precision metric applies, since this is constrained text generation, not prediction. Success is measured as the percentage of generated messages that pass the rules-based content filter unmodified; target $\geq 99\%$ on the MVP test set. No fabricated accuracy figures are claimed.

AI-006 — Failure Handling: If the LLM call fails, times out, or its output fails the content filter, the system falls back automatically to a pre-written, non-AI template with the same data substituted in. All fallbacks are logged.

AI-007 — Human Oversight: The coordinator can review and edit the weekly digest before relying on it. No AI-generated text can alter a calendar entry — calendar mutations only ever originate from a staff action (Section 5, Module 1-3).

12. Database Requirements

Entity	Purpose	Key Attributes	PK	FK
Facility	A clinic/hospital using the system	name, address	facility_id	—
StaffUser	Doctor/Coordinator/Admin account	name, role, email, password_hash	user_id	facility_id
Patient	Discharged mother	name, contact_number, pre-ferred_language, delivery_date, discharge_date	patient_id	facility_id
Newborn	Linked infant record	dob	newborn_id	patient_id
DischargePlan	Container for a patient's visit schedule	created_by, created_at, status	plan_id	patient_id, created_by→user_id
RiskFlag	Applied risk condition	flag_type, applied_at	flag_id	plan_id
PostnatalVisit	Individual scheduled visit	visit_type, due_date, status, completed_date, note	visit_id	plan_id
FamilyMember	Consented secondary contact	name, relation, contact_number, pre-ferred_language	family_id	patient_id
ConsentRecord	Consent state history	consent_given, recorded_by, timestamp	consent_id	family_id, recorded_by→user_id
NudgeLog	Record of each message attempt	channel, recipient_type, sent_at, status	nudge_id	visit_id
AuditLog	Immutable event history	actor_id, action, entity, timestamp	audit_id	—

Constraints: `PostnatalVisit.status` $\in$ {`upcoming`, `due_today`, `overdue`, `completed`, `missed`}. `ConsentRecord.consent_given` defaults false; a nudge to a `FamilyMember` is blocked at the API layer unless the latest `ConsentRecord` is true.

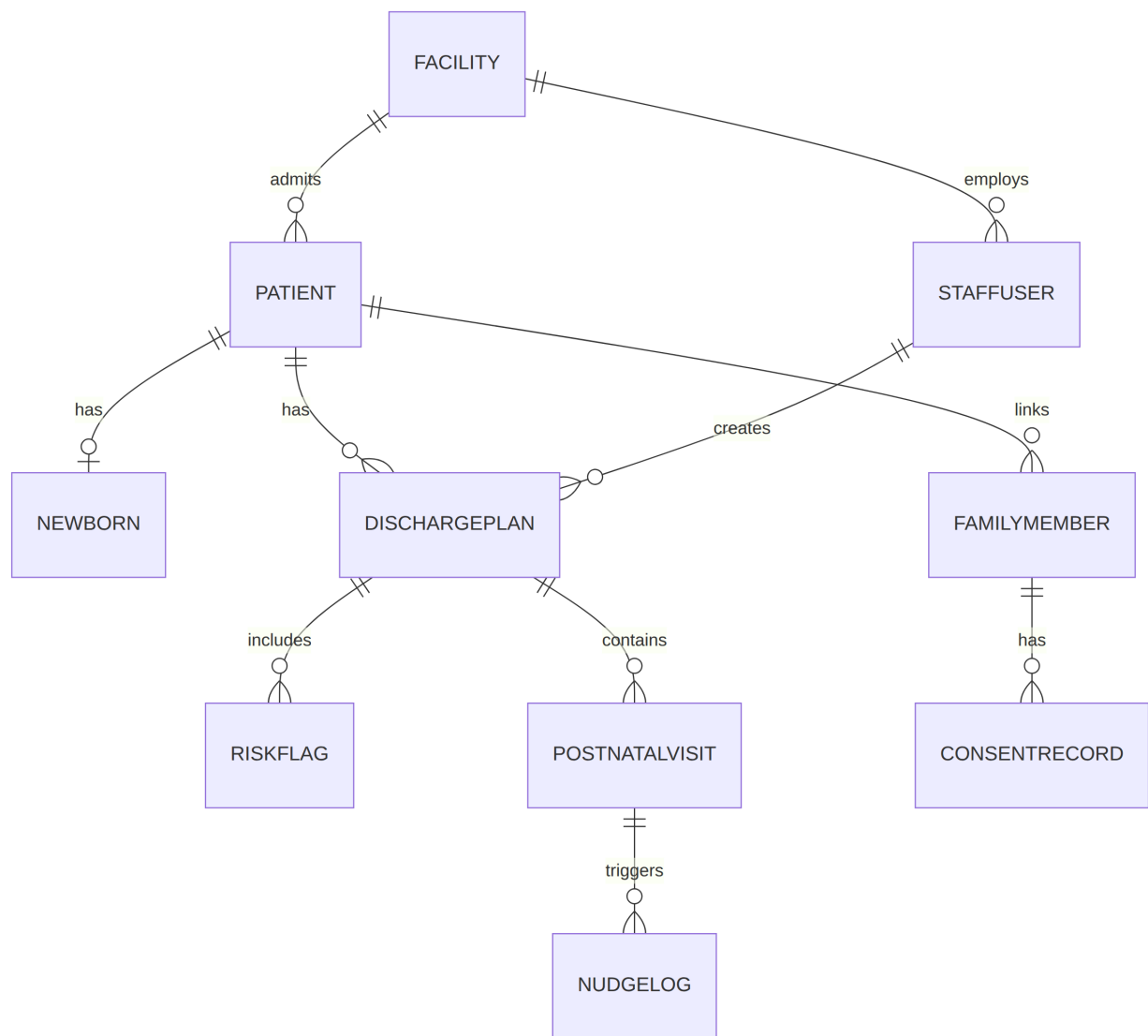


Figure 6: Entity-Relationship Diagram: Core Data Model

13. API Requirements

API ID	Method	Endpoint	Purpose	Auth
API-01	POST	/api/v1/auth/login	Staff login	None (issues JWT)
API-02	POST	/api/v1/patients	Create patient + discharge plan	JWT (Doctor)
API-03	GET	/api/v1/patients/{id}	Patient detail incl. visits/consent	JWT
API-04	POST	/api/v1/patients/{id}/risk-flags	Apply risk override	JWT (Doctor)
API-05	GET	/api/v1/worklist	Ranked overdue worklist for facility	JWT (Coordinator)
API-06	POST	/api/v1/visits/{visit_id}/complete	Mark visit complete	JWT (Coordinator)
API-07	POST	/api/v1/patients/{id}/family-members	Add family member + consent	JWT (Coordinator)
API-08	PATCH	/api/v1/family-members/{id}/consent	Revoke/grant consent	JWT (Coordinator)
API-09	POST	/api/v1/webhooks/twilio/status	Delivery-status callback	Twilio signature
API-10	GET	/api/v1/reports/outcomes	Due-vs-completed report	JWT (Admin/Doctor)
API-11	GET	/api/v1/audit-log/{patient_id}	Audit trail for a patient	JWT (Admin)

Example — API-06 request/response:

POST /api/v1/visits/42/complete

```
{ "completed_date": "2026-09-10", "note": "seen at home visit" }
```

Response 200:

```
{ "visit_id": 42, "status": "completed", "audit_id": 1091 }
```

14. External System Integrations

System	Purpose	Data Exchanged	Auth	Failure Handling
Twilio WhatsApp Business API (sandbox for demo)	Nudge delivery	Phone number, message text	API key	Falls back to SMS; logs failure
Twilio SMS	Nudge fallback	Phone number, message text	API key	Failure flagged on worklist
Multilingual LLM API (optional, e.g., if a platform partner is provided)	Localize digest/nudge text	Structured due/overdue data only, no free-text PII	API key	Falls back to static template (AI-006)

15. UI/UX Requirements

Only staff-facing screens exist; the mother and family interact solely through WhatsApp/SMS.

Screen	Purpose	Key Components
Login	Staff authentication	Email/password, JWT issuance
Facility Worklist (Dashboard)	Coordinator's daily view	Ranked overdue list, quick "mark complete" action
New Discharge Plan	Doctor creates a plan	Patient fields, default visit dates, risk-flag selector
Patient Detail	Full visit/consent history	Visit timeline, family member list, consent toggle
Outcomes Report	Admin/doctor reporting view	Date range filter, completion-rate chart, CSV export
Staff Management	Admin only	Add/deactivate staff, assign roles

User journey: Login → Facility Worklist → (New Discharge Plan | Patient Detail) → Mark Complete / Manage Consent → Outcomes Report.

16. Error Handling

Error ID	Condition	System Response	User Message	Recovery
ERR-01	Missing contact number on discharge form	Block save	"Contact number is required"	User re-enters
ERR-02	Invalid/expired JWT	401 response	"Session expired, please log in again"	Re-login
ERR-03	Twilio send failure	Retry via SMS	(staff-facing) "WhatsApp failed, SMS attempted"	Auto-fallback
ERR-04	Both WhatsApp and SMS fail	Flag on worklist	"Nudge undelivered — contact manually"	Manual coordinator follow-up
ERR-05	LLM call failure/timeout	Fallback to template	None (transparent to recipient)	Static template sent
ERR-06	Content filter rejects AI output	Discard, use template	None	Logged for review
ERR-07	Duplicate patient record	Block creation	"Patient with this contact already exists"	Merge or edit existing
ERR-08	Unauthorized role action (e.g., coordinator editing risk flag)	403 response	"You don't have permission for this action"	Escalate to doctor
ERR-09	Database unavailable	503 response	"System temporarily unavailable"	Retry with backoff

17. Security Requirements

- JWT-based authentication with short-lived access tokens and refresh tokens.
 - RBAC enforced server-side for every endpoint (Doctor / Coordinator / Admin scopes).
 - All API traffic over HTTPS/TLS; Twilio webhook payloads verified by signature.
 - Passwords hashed (bcrypt/argon2); no plaintext storage.
 - Rate limiting on nudge-trigger and login endpoints to prevent abuse.
 - Consent is a hard gate: the API rejects any nudge-send attempt to a FamilyMember without an active ConsentRecord.
 - Audit log is append-only (no update/delete permission at the database role level).
 - No PII (names, phone numbers) is written to application logs — only internal IDs.
-

18. Data Requirements

- **Data sources:** Manually entered by doctor/coordinator only; no OCR, no external EHR feed for MVP.
 - **Data formats:** Structured JSON via API; Postgres relational storage.
 - **Validation:** Server-side validation on all mandatory fields (contact number format, date logic — due date must postdate delivery date).
 - **Retention:** Patient records retained per facility policy (configurable); audit log retained indefinitely for MVP.
 - **Privacy:** All demo and pilot data for the hackathon is synthetic and fully anonymised; no real patient data is used at any stage of development or demonstration.
 - **Backup/Recovery:** Daily automated Postgres backup (standard managed-database feature) for any post-hackathon pilot deployment.
-

19. Assumptions & Constraints

Assumptions: the facility has (or can obtain) a Twilio sandbox account; staff have access to a computer/tablet for the dashboard; the mother or a household member has a WhatsApp-reachable phone number (SMS fallback covers basic-phone cases).

Constraints: 4-5 week hackathon build sprint; no EHR integration; no real patient data; must remain strictly non-clinical (no diagnostic, dosage, or treatment-pathway logic anywhere in the system).

Dependencies: Twilio API availability; optional multilingual LLM API if provided by hackathon organizers; Postgres and Redis hosting.

20. Requirement Traceability Matrix

Req ID	Requirement	Module	User Story	Use Case	Test Case
FR-001	Create discharge plan	M1	US-001	UC-01	TC-01
FR-002	Risk-based interval override	M1	US-002	UC-01	TC-02
FR-005	Compute visit status	M2	US-003	UC-03	TC-03
FR-006	Record visit completion	M2	US-004	UC-02	TC-04
FR-008	View ranked worklist	M3	US-003	UC-02	TC-05
FR-011	Trigger scheduled nudge	M4	US-005	UC-03	TC-06
FR-016	Add family member + consent	M5	US-006	UC-04	TC-07
FR-017	Revoke consent	M5	US-008	UC-04	TC-08
FR-019	Due-vs-completed report	M6	US-007	UC-05	TC-09
FR-021	Immutable audit log	M7	—	UC-02	TC-10

21. Testing Requirements

Covers unit (per-module business logic), integration (API ↔ Postgres ↔ scheduler), system (full discharge-to-nudge cycle), API (contract tests per endpoint in Section 13), UI (staff dashboard flows), security (RBAC boundary tests, consent-gate tests), AI/ML (content-filter pass-rate on generated messages), and user acceptance (doctor/coordinator walkthrough with synthetic patients).

Test ID	Req ID	Scenario	Input	Expected Result	Priority
TC-01	FR-001	Create plan with valid data	Complete discharge form	4 visit records created	Must
TC-02	FR-002	Apply hypertension risk flag	Risk flag = hypertension	Weekly interval applied	Must
TC-03	FR-005	Visit crosses due date	Visit due_date = today	Status → due_today	Must
TC-04	FR-006	Mark visit complete	visit_id, completion date	Status → completed, audit entry created	Must
TC-05	FR-008	Load worklist with 3 overdue patients	Facility ID	List sorted by days-overdue descending	Must
TC-06	FR-011	Nudge trigger on overdue visit	Overdue visit, consent = true	WhatsApp message sent, NudgeLog entry created	Must
TC-07	FR-016	Add family member with consent=true	Family member details	Nudges enabled for that contact	Must
TC-08	FR-017	Revoke consent mid-cycle	family_id	Subsequent nudges to that contact blocked	Must
TC-09	FR-019	Generate report for date range	Start/end dates	Correct completion percentages returned	Should
TC-10	ERR-06	AI output contains disallowed clinical term	Simulated bad LLM output	Filter rejects, template fallback sent	Must

22. Deployment Requirements

- **Development:** local Docker Compose (API + Postgres + Redis).
- **Testing/Demo:** single cloud VM or PaaS instance (e.g., Render/Railway) with managed Postgres; Twilio sandbox credentials.
- **Production (future pilot):** managed Postgres with backups, containerized API with auto-restart, environment-separated Twilio credentials.
- **Environment variables:** DATABASE_URL, TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, JWT_SECRET, LLM_API_KEY (optional).
- **CI/CD:** basic GitHub Actions pipeline (lint + test) is a Should, not a Must, for the hackathon timeline.

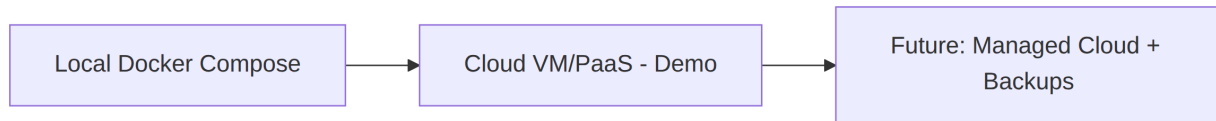


Figure 7: Deployment Architecture: Environments

23. Monitoring & Logging

- Application logs (request ID, endpoint, status code — no PII).
 - Scheduled-job logs (daily status-computation run, nudge-batch run, success/failure counts).
 - Twilio delivery-status metrics (sent/delivered/failed counts per day).
 - Alert threshold: if failed nudges exceed 10% of attempts in a day, flag for coordinator review.
-

24. MVP Requirements for Hackathon

Must Build: discharge plan creation (FR-001, FR-002); calendar status computation (FR-005); visit completion (FR-006); coordinator worklist (FR-008); WhatsApp/SMS nudge sending via Twilio sandbox (FR-011); consent capture and gating (FR-016, FR-017); content filter and template fallback (AI-006); audit log (FR-021).

Should Build: weekly digest generation (FR-012); outcomes report (FR-019); risk-flag UI polish (FR-003, FR-004).

Could Build: multilingual nudge generation (AI-002/003 in a non-English language); two-way reply capture (FR-015); anonymised export (FR-020).

Future: RCH/Kilkari data interoperability; multi-facility rollout; native mobile app; predischARGE-summary OCR ingestion.

25. Future Enhancements

Post-hackathon: expand the calendar to the full antenatal period (not just postnatal); support multi-facility deployments with facility-level admin roles already scaffolded in Section 12; explore an opt-in data-sharing pathway with RCH once government API access is available; add IVR fallback for households without WhatsApp access, mirroring Kilkari's channel mix.

26. Risks

Risk	Probability	Impact	Mitigation
WhatsApp Business API production approval delays	Medium	Medium	Use Twilio sandbox for demo/pilot; do not block MVP on production approval
Mother/family phone number unreachable or stale	Medium	High	SMS fallback; worklist flags unreachable contacts for manual follow-up
Clinical-line creep in AI-generated text	Low	High	Rules-based content filter (AI-006) with template fallback; no free-text clinical fields anywhere in the schema
Low staff adoption at busy facilities	Medium	Medium	Keep discharge-plan entry under 2 minutes; no EHR integration required
Consent friction / family reluctant to share access	Medium	Low	Consent is explicit, granular, and revocable at any time (FR-017)
No real patient data available for testing	High	Low	Synthetic data generator built with team doctor; explicitly labeled synthetic in all demo material
Scheduler job failure causing missed nudges	Low	Medium	Automated retry (FR-005 exception handling) plus daily job-failure alerting

27. Final Requirements Summary

ID	Requirement	Type	Priority	Module
FR-001	Create discharge plan	Functional	Must	M1
FR-002	Risk-based interval override	Functional	Must	M1
FR-005	Compute visit status	Functional	Must	M2
FR-006	Record visit completion	Functional	Must	M2
FR-008	View ranked worklist	Functional	Must	M3
FR-011	Trigger scheduled nudge	Functional	Must	M4
FR-016	Add family member + consent	Functional	Must	M5
FR-017	Revoke consent	Functional	Must	M5
FR-019	Due-vs-completed report	Functional	Should	M6
FR-021	Immutable audit log	Functional	Must	M7
AI-006	Content-filter fallback	AI/ML	Must	M4
NFR-SEC	RBAC + consent gating	Non-Functional	Must	Cross-cutting

28. Hackathon Implementation Blueprint

Folder Structure

```
cara/  
  backend/  
    app/  
    models/      # Patient, DischargePlan, PostnatalVisit, FamilyMember, ConsentRecord, NudgeLog,  
    routers/     # auth, patients, visits, worklist, family, reports, webhooks  
    services/    # calendar_engine.py, nudge_engine.py, content_filter.py  
    scheduler/   # daily status job, nudge batch job  
    tests/  
  frontend/  
    src/  
    pages/       # Login, Worklist, NewDischargePlan, PatientDetail, OutcomesReport, StaffAdmin  
    components/  
  docker-compose.yml
```

Backend Modules: Auth, Discharge Planning, Calendar Engine, Worklist, Nudge Engine (+ Content Filter), Consent, Reporting, Audit.

Frontend Modules: Login, Worklist Dashboard, Discharge Plan Form, Patient Detail, Outcomes Report, Staff Admin.

Database Tables: Facility, StaffUser, Patient, Newborn, DischargePlan, RiskFlag, PostnatalVisit, FamilyMember, ConsentRecord, NudgeLog, AuditLog.

APIs: as listed in Section 13.

AI/ML Components: optional multilingual generation service + rules-based content filter (Section 11).

External Services: Twilio (WhatsApp sandbox + SMS), optional multilingual LLM API.

Deployment: Docker Compose locally → single cloud VM/PaaS for demo.

Team Responsibilities: Doctor — define default visit intervals and risk-based override rules, validate synthetic patient scenarios, review demo script for clinical accuracy of non-clinical framing. Tech members — split across backend (calendar engine + nudge engine), frontend (dashboard), and integrations (Twilio, optional LLM).

Development Order (5-week sprint): Week 1 — data model + auth + discharge-plan creation (FR-001/002). Week 2 — calendar engine + worklist (FR-005/008). Week 3 — Twilio nudge integration + consent gating (FR-011/016/017). Week 4 — content filter, outcomes report, audit log (AI-006, FR-019/021). Week 5 — synthetic data polish, demo video, mentor rehearsal.

5 Most Critical Requirements for Hackathon Success

1. **FR-001 (Create Discharge Plan)** — without this, no calendar exists; it's the entry point for the entire demo.
2. **FR-005 (Compute Visit Status)** — the core logic that makes the system “agentic” rather than a static list.
3. **FR-011 (Trigger Scheduled Nudge)** — the visible, demoable proof that follow-through is automated.
4. **FR-016/FR-017 (Consent Capture and Revocation)** — the guardrail that makes the family-inclusion feature trustworthy and DPDP-appropriate.

5. **AI-006 (Content-Filter Fallback)** — the single requirement that keeps the system inside the hackathon’s non-clinical eligibility rule under all conditions, including AI failure.